

ATTORNEY SHIELD

ASI 2.0 - Native Android & iOS

ASI 2.0 Testing Plan

Test strategy for kotlin & swift

Draft for approval - 2026-08-10

Colour system: Attorney-Shield Color Scheme Review

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DRAFT FOR APPROVAL

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Date: 2026-08-10 **Applies to:** kotlin (Android) and swift (iOS) **Companion:** [development plan](#)

CLAUDE.md mandates: **unit tests for every screen**, verification on the Android Emulator / iOS Simulator / Claude Browser against `member-client`, and **test reports saved as journal entries**. This plan makes those concrete.

1. Layers

Layer	Scope	Runs
L1 Unit	View models, API mapping, formatters, token math	Every commit
L2 UI (component)	Each screen renders & reacts to state	Every commit
L3 Integration	Real dev backend: login -> tiles -> call setup	Per phase
L4 Device	Emulator + Simulator, real permissions, real video	Per phase
L5 Parity	Side-by-side vs <code>member-client</code> in Claude Browser	Per phase

Toolchain

- Android: JUnit5 + MockK + Turbine (Flow), Compose UI tests, Robolectric, MockWebServer. `./gradlew test connectedAndroidTest`
- iOS: Swift Testing (or XCTest) + URLProtocol stubs, SwiftUI ViewInspector or snapshot tests. `xcodebuild test`

2. Design-system gates (automated, not eyeballed)

These are the tests that keep the palette honest. All derived from [design/color-system.md](#).

- T-DS-1 Token contrast.** Assert computed WCAG ratios for every foreground/background pair the app actually uses, against the §5 table. Must include the three known traps as **negative** assertions: white-on-Justice-Gold (3.13), white-on-Active-Gold (2.22), Mid-Navy-on-Shield-Navy (1.49) - the test proves the app never forms these pairs.
- T-DS-2 No raw hex.** Lint/test failing on any hex literal in UI code outside the token file.
- T-DS-3 Palette allowlist.** Every colour in the token file is one of the 11 approved hexes; no forbidden hex (`#FF4500`, `#7B2FBE`, `#00B4D8`, `#F0F0F0`) appears anywhere in either repo.
- T-DS-4 CTA foreground.** Primary buttons use `ctaFg` (navy), never white.
- T-DS-5 Dynamic type.** Screens render without clipping at the largest OS font setting.

3. Per-screen unit/UI tests

Login

ID	Case	Expected
T-L-1	Valid credentials	Token stored, navigates to Home
T-L-2	Invalid credentials	GraphQL errors[] surfaced as readable message
T-L-3	user query fails	Login still succeeds; org falls back to DEV_DEFAULTS
T-L-4	casesByUser fails	Login still succeeds; jurisdiction falls back
T-L-5	Email whitespace	Trimmed before send
T-L-6	Network offline	Readable error, no crash, retry available
T-L-7	Password field	Masked; excluded from logs and crash reports

Home

ID	Case	Expected
T-H-1	Tiles load	Sorted by sortOrder
T-H-2	Multilingual labels	English -> org default -> first -> humanized code
T-H-3	Type with no translations	Humanized code (e.g. traffic_stop -> "Traffic Stop")
T-H-4	Missing iconFilePath	Falls back to the code's icon, generic shield if unknown
T-H-5	partnerId null	Chip row hidden entirely
T-H-6	partnerAttorneys fails	Empty list, no error shown, auto-match only
T-H-7	Attorney selected	Request carries attorneyId, not queueId
T-H-8	Auto-match selected	Request carries queueId, not attorneyId
T-H-9	Zero types configured	Empty-state message, no crash
T-H-10	Greeting name	displayName first word -> else email local-part, capitalized
T-H-11	Loading	Skeletons, not a blank screen

Call - the phase machine

Cover all six phases of starting | connecting | live | ended | no-attorney | error.

ID	Case	Expected
T-C-1	Happy path	starting -> connecting -> live
T-C-2	409 from member-call	no-attorney, retryable message, no crash
T-C-3	500	error with status surfaced
T-C-4	Response missing token/apiKey/sessionId	error, never attempts connect
T-C-5	Vonage connect error	error, message surfaced
T-C-6	Publish error	error
T-C-7	Remote hangs up (connectionDestroyed)	ended
T-C-8	sessionDisconnected while live	ended; when not live, phase unchanged
T-C-9	Member hangs up	Teardown; unpublish -> destroy -> disconnect, each guarded
T-C-10	Leave screen mid-call	Full teardown, no leaked publisher/session
T-C-11	Camera/mic denied	Clear explanation + settings route; no crash
T-C-12	Location denied/unavailable/timeout	Call proceeds without coords
T-C-13	Location granted	memberLat/memberLng(/accuracy) included
T-C-14	Backgrounded during call	Documented, deliberate behaviour
T-C-15	ttlSeconds	Sent as 3600

Deep-link return from web checkout

Load-bearing: registration cannot continue without it, and it fires **after the member has paid**, so a failure here strands them mid-signup.

ID	Case	Expected
T-D-1	Universal/App Link, app installed & warm	Opens app, resumes at screen 08
T-D-2	Same link, app cold	Same destination; no lost state
T-D-3	Link carries email	Email pre-filled
T-D-4	Malformed/partial link	Graceful landing, never a crash or blank screen
T-D-5	App not installed	OS fallback behaves; store route intact
T-D-6	"Open app manually" fallback	Works when the automatic link does not
T-D-7	Link replayed / opened twice	Idempotent - no duplicate registration
T-D-8	Tampered or foreign-origin link	Rejected; only attorney-shield.com honoured

T-D-8 matters: a deep link is untrusted input. Nothing in it should be trusted to grant entitlement - only the backend decides what the member has paid for.

4. Integration (L3)

Against the dev backend, using DEV_DEFAULTS seed IDs:

- Real login -> real token; authenticated GraphQL carries Bearer
- Unauthenticated /query is rejected as expected
- member-call returns all five credential fields
- 409 reproduced deliberately (no available attorney) and handled

5. Device verification (L4)

Per phase, both platforms:

- 1 Cold launch -> login -> home -> place a call -> connect -> hang up
- 2 Deny camera, then mic, then location - verify each degradation
- 3 Airplane mode at each step
- 4 Rotation and largest font size on every screen
- 5 Screen reader pass: every control has a meaningful label

Simulator work uses the iOS Simulator tooling; Android uses the emulator. Capture screenshots into the journal entry.

6. Parity vs member-client (L5)

Run the reference locally:

```
cd /Users/samsonphillip/attorney/ASI_2/member-client && npm install && npm run dev
```

Open it in Claude Browser beside the emulator/simulator and diff **behaviour**: greeting text, tile order and labels, chip visibility, routing choice, each call phase's copy, and the 409 message.

Compare behaviour, not pixels. member-client is intentionally off-palette (blue/violet); a visual diff is expected and is *not* a defect. Colour is judged against the PDF, never against the reference app.

member-client has no test runner in package.json and is read-only - we add no tests there.

7. Definition of done (per screen)

- L1 + L2 tests green, including every relevant row above
- Design-system gates (§2) green
- Verified on emulator and simulator

- Parity checked against `member-client`
- Accessibility: labels, dynamic type, 4.5:1 text / 3:1 UI
- Journal entry with results, coverage notes, and screenshots

8. Coverage targets

Area	Target
View models / state logic	>=85% branch
API mapping + error paths	100% of documented error branches
Call phase machine	100% of the six phases
UI composables/views	Every screen has render + state tests

Treat coverage as a floor, not a goal - T-C-2 (409) and T-C-12 (location degradation) matter more than any percentage.

9. Reporting

One journal entry per test cycle at `journals/YYYY-MM-DD-<task>-test-report.md`: what ran, pass/fail counts, failures with root cause, coverage, screenshots, and open defects. Written as work happens, not retrofitted.

10. Scope note - what these tests do and do not cover

The design board is **66 frames (57 native, 9 web)**, not the 35 its masthead claims. The test IDs above cover the **unblocked spine** - welcome, login, home, call, and the deep-link return. That is Phases 0-2.

The 9 web frames are out of scope for these tests. Plan choice, Stripe checkout and confirmation are web pages; we test the *handoff back* (§3 deep-link), not the checkout itself.

Not yet assigned test IDs, deliberately, because they have no endpoints (development plan §3):

Surface	Screens
Native registration	08-12
PIN / end session	12, 34
Document vault	14, 14A-14D, 31
Situations, activity, family, plan	13B, 13C, 27B, 32, 33A-33D
Nudges & notifications	15, 22-26
Trial gate & conversion	V2, T5-T8

Surface	Screens
Guest gates	G1-G3
Grace / expired states	35

When those unblock, each needs the same treatment as §3: happy path, every error branch, and the degradation cases.

11. Gaps this plan cannot yet close

- **Store compliance is untestable by us.** Whether web-purchased subscriptions plus a deep-link return passes App Store / Play review is a policy question, not a test case (development plan §8). It needs an answer before Phase 3, because a rejection invalidates the flow rather than delaying it.
- **Family capacity is ambiguous in the reference** - stated four different ways (development plan §9.6). No assertions can be written for the member stepper until "up to 5" is confirmed as inclusive or exclusive of the primary account.
- **Token expiry has no defined correct behaviour** - no refresh operation exists (development plan §8). Tests will assert whatever we decide; the decision comes first.
- **Vonage native SDK behaviour is unproven** - the Phase 0 spike must land before Call tests can be trusted.